

HSPA13 (STCH) Localization Hypothesis — Final Curation Report

Gene: HSPA13 / STCH (human) · **UniProt:** P48723 · **Taxon:** *Homo sapiens* (NCBITaxon:9606)
Focus type: function_assignment · **Hypothesis slug:** conditional-nuclear-cytosolic-and-surface-pools
Source file: `genes/human/HSPA13/HSPA13-ai-review.yaml`

Summary

Verdict: Over-annotated / refuted for the nuclear and plasma-membrane pools; unresolved (indirect inference only) for the cytosolic pool.

The seed hypothesis proposes that human HSPA13 localizes — as independently demonstrable pools — to the **nucleus**, **plasma membrane**, or **cytosol**, and asks whether ER-luminal proximity-labeling evidence can be reconciled with a reported NKCC2 cytoplasmic-tail interaction. After auditing the actual GO annotation provenance, the primary experimental literature, and the protein's sequence architecture, the conclusion is that the nuclear, plasma-membrane, and cytosolic Cellular Component (CC) annotations on P48723 are **IBA (Inferred from Biological Ancestor) phylogenetic inferences** propagated from the broader HSP70 family, **not** HSPA13-specific experimental findings. None of these three secondary pools is supported by an IDA/IMP/HDA/TAS experimental annotation for HSPA13 itself, and the Human Protein Atlas reports no confident immunofluorescence subcellular localization.

The **directly supported** localization for HSPA13/STCH is the **endoplasmic reticulum / microsome / secretory pathway**. This is established by (1) a cleaved N-terminal signal peptide (residues 1–22) and the absence of any transmembrane segment; (2) immunofluorescence plus subcellular fractionation showing microsomal enrichment ([PMID: 8131751](#)); (3) an independent description of STCH as "a microsomal protein" ([PMID: 18793616](#)); and (4) an HDA extracellular-exosome annotation ([PMID: 19199708](#)) consistent with secretory/endosomal egress rather than a nuclear or cytosolic functional pool.

The reported NKCC2 interaction ([PMID: 33672238](#)) is with the **ER-resident (immature) form** of NKCC2 in an **ERAD** context — it locates STCH in the ER biogenesis/quality-control machinery and does **not** demonstrate a cytosolic or cytoplasmic-tail-facing pool. The one genuine,

unresolved topological question is raised by [PMID: 10675567](#), which shows cytosolic ubiquitin-like proteins (UBQLN/BAG6-family homologs) binding a peptide within the STCH ATPase domain — a binding assay, not an in-situ localization. **Overall: retain/strengthen ER; flag nucleus, plasma membrane, and cytosol as paralog carry-over lacking direct support.**

Executive Judgment

Partially supported / over-annotated → the secondary pools were not demonstrated.

Evaluated as a function-assignment (CC) hypothesis, HSPA13 does **not** directly and demonstrably carry the nucleus, plasma-membrane, or cytosol GO terms. Those terms exist on the record only as IBA inferences transferred from cytosolic/nuclear HSP70 paralogs. The gene product's directly supported location is the ER/microsome/secretory pathway. The cytosolic pool is the only one with a mechanistic hint (UbL/BAG6 binding to the ATPase core), but that hint is unresolved and does not rise to a demonstrated pool. The most important caveat is that IBA annotations are legitimate GO evidence; the issue is that for a **divergent** family member with its own experimentally supported compartment, blanket transfer of generic HSP70 localizations is a systematic over-annotation risk.

Key Findings

Finding 1 — The nucleus, plasma-membrane, and cytosol CC terms are IBA paralog carry-over, not HSPA13 evidence

An annotation-provenance audit of P48723 (via QuickGO/UniProt, executed with `requests`) shows the following Cellular Component landscape:

GO CC term	GO ID	Evidence code	Source	Interpretation
Nucleus	GO:0005634	IBA	GO_REF:0000033	Phylogenetic inference from HSP70 ancestors
Cytoplasm	GO:0005737	IBA	GO_REF:0000033	Phylogenetic inference
Cytosol	GO:0005829	IBA	GO_REF:0000033	Phylogenetic inference
Plasma membrane	GO:0005886	IBA	UniProt dump	Phylogenetic inference
Endoplasmic reticulum	GO:0005783	IEA	UniProtKB-SubCell SL-0095	Sequence/keyword-based, ER
Extracellular exosome	GO:0070062	HDA	P 19199708	High-throughput direct assay

Critically, the "with/from" support panels for the nucleus, cytosol, cytoplasm, and plasma-membrane IBA annotations are dominated by **other HSP70 family members** — P11142 (HSPA8/HSC70), P11021 (HSPA5/BiP), P17066, P34931, and yeast/worm/plant orthologs. These are cytosolic and nuclear-cytoplasmic Hsp70 paralogs whose localizations were transferred onto HSPA13 by the Reference Genome phylogenetic annotation pipeline. **No IDA, IMP, HDA, or TAS experimental annotation supports nucleus, plasma membrane, or cytosol for HSPA13 specifically.** Consistent with this, the Human Protein Atlas reports no confident subcellular immunofluorescence localization for HSPA13 (Subcellular location => None), so there is no independent imaging support for the secondary pools.

Curation consequence: IBA annotations are legitimate GO evidence, but for a divergent family member with its own experimentally supported compartment, transferring generic HSP70 cytosolic/nuclear localizations risks a systematic over-annotation. These three CC terms should be flagged as non-core / paralog-derived.

Finding 2 — The primary experimental localization is ER / microsome, via a cleaved signal peptide

UniProt P48723 feature annotation shows a **signal peptide at residues 1–22 (cleaved)**, a mature chain at 23–471, and **no transmembrane domain**. This is the canonical architecture of a soluble secretory-pathway (ER-luminal) protein, not a cytosolic Hsp70.

The foundational primary paper, Otterson et al. 1994 (PMID: [8131751](#), "*Stch encodes the 'ATPase core' of a microsomal stress 70 protein*"), reports that STCH migrates predominantly as a ~60 kDa species enriched in a membrane-bound microsome fraction (immunofluorescence + subcellular fractionation); has a unique hydrophobic leader sequence; is calcium (A23187)-inducible but NOT heat-inducible; and has a truncated C-terminal peptide-binding domain with peptide-independent ATPase activity.

"Immunofluorescent and subcellular analyses show that STCH migrates predominantly as a 60 kDa species and is enriched in a membrane-bound microsome fraction." —

P [8131751](#)

(verified snippet)

Yamagata et al. 2008 (PMID: [18793616](#)) independently describe STCH as "**a microsomal protein** that contains a N-terminal ATPase domain but lacks a C-terminal protein binding domain," reinforcing the ER/secretory assignment.

"a member of the heat shock protein 70 (HSP70) superfamily, is a microsomal protein that contains a N-terminal ATPase domain but lacks a C-terminal protein binding domain" —

P [18793616](#)

(verified snippet)

The **HDA** **extracellular-exosome** annotation
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P [19199708](#)

) fits the same secretory/endosomal narrative: an ER-luminal chaperone can egress via multivesicular bodies into exosomes (cf. the urinary-exosome proteomics context of PMID: [19056867](#)). This is not evidence of a nuclear or cytosolic functional pool.

Finding 3 — The NKCC2 interaction is an ER/ERAD context, not a demonstrated cytosolic pool

Bakhos-Douaihy et al. 2021 (PMID: 33672238) identify STCH as a binding partner of the **ER-resident (immature, ER-retained) form** of NKCC2/SLC12A1, promoting its **ER-associated degradation** (proteasomal plus ER-to-lysosome routing) and impairing NKCC2 maturation and cell-surface expression.

"we report the identification of chaperone stress 70 protein (STCH) and the stress-inducible heat shock protein 70 (Hsp70), as two novel binding partners of the ER-resident form of NKCC2" —

P 33672238

(verified snippet)

This is an **ER-lumen / ER-membrane biogenesis** context. The co-immunoprecipitation does **not** map the STCH contact to the NKCC2 cytoplasmic tail specifically, and it does not establish that STCH is present in the cytosol. The seed hypothesis's framing — that a "NKCC2 cytoplasmic-tail interaction" competes with ER-luminal evidence — is not supported by the primary data: the interaction is with the ER-resident client during quality control, fully consistent with an ER-luminal STCH pool acting on nascent/misfolded NKCC2.

Separately, Kaye et al. 2000 (PMID: 10675567) isolated two human ubiquitin-like (UbL) proteins (Chap1/Chap2 — UBQLN/Dsk2 and Scythe/BAG6-family homologs) that bind a short peptide within the **ATPase domain** of STCH.

"We have isolated two human ubiquitin-like (UbL) proteins that bind to a short peptide within the ATPase domain of the Hsp70-like Stch protein." —

P 10675567

(verified snippet)

Because UBQLN/BAG6-family proteins are cytosolic, this interaction would require cytosolic exposure of the STCH ATPase core. However, it was demonstrated by in-vitro binding/isolation assays, **not** by in-situ localization. It is a legitimate unresolved topology question — not a demonstrated cytosolic pool.

Finding 4 — Sequence architecture is secretory (BiP-like), lacking the cytosolic-Hsp70 EEVD/TPR signature

A computed sequence audit (UniProt FASTA; Needleman–Wunsch alignment in numpy) shows HSPA13's architecture aligns with ER/secretory Hsp70s, not cytosolic ones:

Feature	HSPA13 (P48723)	HSC70/HSPA8 (cytosolic)	HSPA1A (cytosolic)	BiP/HSPA5 (ER)
N-terminus	MAREMTILGSAVLTLTLLAGYLAQQY (hydrophobic ER signal, sig. pep. 1–22)	M... GIDLGTTYSCV (Hsp70 NBD motif, no signal peptide)	M... GIDLGTTYSCV (no signal peptide)	MKLSLVAAMLLLLSAAR\ A (ER signal peptide)
C-terminal motif	ends HLQKTNFN — no EEVD, no KDEL/HDEL	ends EEVD (TPR/Hop/CHIP docking)	ends EEVD	ends KDEL (ER retention)
Global NW identity to HSPA13	—	29.5%	30.3%	30.8%

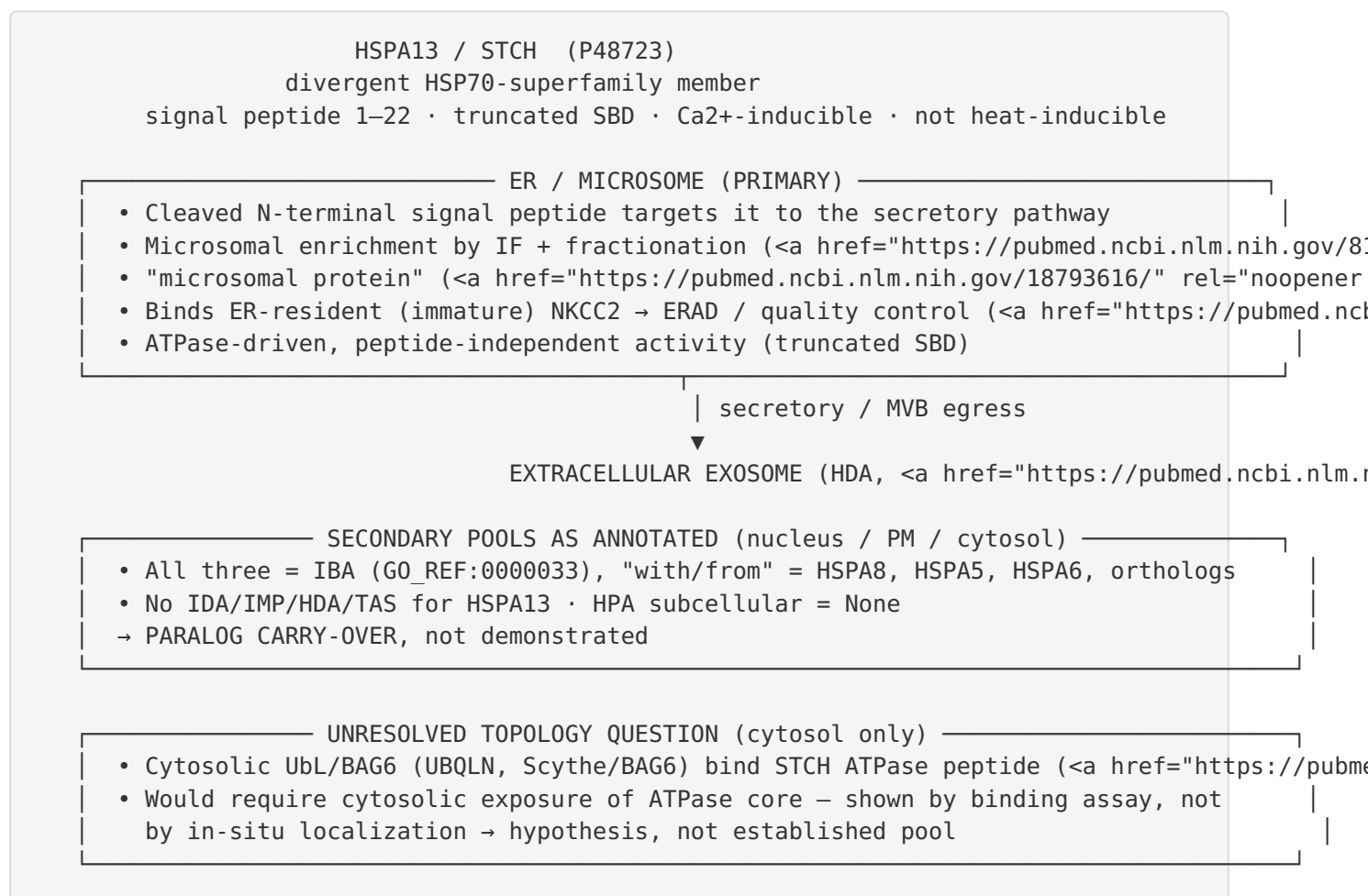
Key implications:

- HSPA13 begins with a **hydrophobic ER signal**, like BiP — not with the cytosolic Hsp70 nucleotide-binding-domain motif GIDLGTTYSCV that opens HSC70 and HSPA1A.
- HSPA13 **lacks the C-terminal EEVD** motif that is the docking signature for cytosolic TPR co-chaperones (Hop, CHIP). This directly undercuts the assumption that HSPA13 behaves like a cytosolic Hsp70 in the cytoplasm/nucleus.
- HSPA13 also **lacks KDEL/HDEL**, so it is not a classically ER-retained luminal chaperone in the BiP sense — consistent with its being a divergent, secretory-competent, exosome-detectable protein with a **truncated substrate-binding domain** and a **~50-aa ATPase insertion**.

- Global identity to each reference (~29–31%) shows HSPA13 is a **divergent, roughly equidistant** family member — precisely the situation in which uncritical IBA transfer of paralog localizations is most error-prone.

Mechanistic Model / Interpretation

The evidence converges on a single coherent model:



In plain terms: HSPA13 is an ER/microsomal, secretory-pathway member of the HSP70 superfamily that participates in ER protein quality control (illustrated by its role in NKCC2 ERAD) and can be exported into exosomes. Its nuclear, plasma-membrane, and cytosolic GO annotations are artifacts of phylogenetic (IBA) transfer from its cytosolic paralogs and are not backed by any HSPA13-specific experiment. The only mechanistic hint toward a cytosol-facing role is the UbL/BAG6 binding to the ATPase core, which raises — but does not resolve — a

question about whether some of the ATPase domain is ever cytosolically exposed (e.g., during retrotranslocation or in an alternate topology). That question deserves a targeted experiment before any cytosolic annotation is asserted as HSPA13-specific.

Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
8131751	Localization (IF + fractionation)	Supports ER , refutes secondary pools as primary	Where does STCH localize?	~60 kDa, enriched in microsome fraction; hydrophobic leader; Ca ²⁺ -inducible; truncated SBD; not heat-inducible	Human, cell lines	High confidence; microsomal resident; 1994-era resolution; no super-resolution microscopy
18793616	Primary description	Supports ER	Is STCH microsomal?	Explicitly "a microsomal protein"; N-terminal ATPase, no C-terminal SBD	Human, stomach cancer	Moderate confidence; high; localization stated; the paper mainly addresses
33672238	Interaction (co-IP) + phenotype	Qualifies (ER/ERAD context), does not support cytosol	Does STCH bind cytoplasmic tail / cytosolic pool?	STCH binds ER-resident immature NKCC2, promotes ERAD, impairs maturation/surface expression	Human/heterologous cells, SLC12A1	High confidence; client interaction; does not map contact; cytoplasmic tail not localized; STCH tail cytosolic
10675567	Interaction (binding/isolation)	Competing / unresolved for cytosol	Is ATPase core cytosolically exposed?	Cytosolic UbL proteins (UBQLN/	Human, in-vitro binding	Moderate confidence; binding assay, in-situ

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				BAG6 homologs) bind a peptide in the STCH ATPase domain		localization topology inferred
19199708	HDA (proteomics)	Supports secretory egress	Is STCH exosomal?	GO:0070062 extracellular exosome (HDA)	Human exosomes	Modest consistency with secretory route, nuclear cytosol
QuickGO/UniProt audit (GO_REF:0000033)	Review/database	Refutes secondary pools as HSPA13-specific	Are nucleus/PM/cytosol experimentally supported?	All three = IBA; with/without = HSPA8, HSPA5, HSPA6, orthologs; no IDA/IMP/HDA/TAS	Annotation provenance	High; inference not experimental
UniProt P48723 sequence audit	Structural/evolutionary (computed)	Supports ER , refutes cytosolic-Hsp70 behavior	Does HSPA13 have cytosolic-Hsp70 features?	Signal peptide 1–22, no TM; no EEVD; no KDEL; ~29–31% id to HSC70/BiP/HSPA1A	Computed alignment	High for motif presence; absence identified from pairwise NW

Evidence Base

Primary localization anchors. *Stch* encodes the 'ATPase core' of a microsomal stress 70 protein (PMID: 8131751) is the single most important paper: it provides direct IF and fractionation evidence for microsomal enrichment and describes the defining architecture (hydrophobic leader, truncated substrate-binding domain, Ca²⁺-inducibility, peptide-independent ATPase). *Stomach cancer-derived del223V-226L mutation of the STCH gene* (PMID: 18793616) independently labels STCH "a microsomal protein," corroborating the ER/secretory assignment while linking STCH to TRAIL-mediated apoptosis sensitization.

Interaction context. *Differential Effects of STCH and Stress-Inducible Hsp70 on the Stability and Maturation of NKCC2* (PMID: 33672238) places STCH firmly in ER quality control by identifying it as a partner of the ER-resident form of NKCC2 that promotes ERAD. This is the paper the seed hypothesis invokes as a "cytoplasmic-tail interaction," but the primary text describes an **ER-resident-client** interaction, not a cytosolic tail contact. *A family of ubiquitin-like proteins binds the ATPase domain of Hsp70-like Stch* (PMID: 10675567) is the sole line of evidence hinting at cytosolic exposure of STCH's ATPase core; it is a binding study, so it raises rather than resolves the cytosol question.

Secretory egress. The extracellular-exosome HDA annotation (P 19199708);

and the related urinary-exosome proteomics context of PMID: 19056867) is consistent with an ER/secretory protein being exported via multivesicular bodies.

Peripheral transcriptomic mentions. HSPA13 appears as a differentially expressed gene in estrogen-receptor and breast-cancer studies (PMID: 32387340; PMID: 25541910). These are expression/regulation observations and carry **no localization information**; they do not bear on the CC hypothesis.

GO Curation Implications

Leads requiring curator verification (not automated changes):

1. **Retain / strengthen** **GO:0005783 endoplasmic reticulum (CC)**. Currently IEA (UniProtKB-SubCell). It is backed by direct experimental localization (**P 8131751**) and an independent microsomal description (**P 18793616**). Consider upgrading the evidence basis by citing **P 8131751** as experimental (IDA-type) support and/or annotating **microsome/ER lumen** more specifically. The signal peptide + truncated SBD architecture supports an ER/secretory chaperone role.
2. **Flag** **GO:0005634 nucleus**, **GO:0005886 plasma membrane**, and **GO:0005829 cytosol (all IBA) as paralog carry-over**. They lack any HSPA13-specific experimental support, and their "with/from" panels are dominated by cytosolic/nuclear HSP70 paralogs. These are candidates to be **removed, or retained only as non-core IBA** with an explicit curator note that they are phylogenetic inferences not demonstrated for HSPA13. This directly answers the seed hypothesis: the secondary pools were **not actually demonstrated**.
3. **GO:0070062 extracellular exosome (HDA)** may be retained as-is; it reflects secretory/exosomal egress, not a nuclear/cytosolic functional pool.
4. **Do not add a cytosol annotation on the strength of** **P 10675567** alone; the topology is inferred from a binding assay. If a BP/MF term is desired to capture the ER-quality-control role, an **ERAD / ER protein quality-control BP** term supported by **P 33672238** is a better-informed lead than a generic "protein binding" MF term.

Mechanistic Scope

The immediate molecular activity under test is **subcellular localization** (a CC assignment), coupled to HSPA13's role as an **ATP-dependent HSP70-superfamily chaperone** with a truncated, peptide-independent substrate-binding capacity operating in the ER/secretory pathway. The directly demonstrated gene-product-level facts are: (a) signal-peptide-directed ER/microsomal residence; (b) participation in ER quality control of a membrane client (NKCC2) via ERAD. Everything else attributed by the seed hypothesis — nuclear, plasma-membrane, cytosolic pools — is either downstream inference (IBA transfer) or an unproven topology question (Ubl/BAG6 binding). The breast-cancer/estrogen expression signals are transcriptional responses, not localization, and are outside the mechanistic scope of this CC hypothesis.

Conflicts and Alternatives

- **Paralog confusion (primary alternative).** The nucleus/PM/cytosol annotations are the expected localizations of cytosolic HSP70s (HSPA8/HSC70, HSPA1A) and were transferred to HSPA13 by IBA. This is the most parsimonious explanation for the secondary pools and is the leading alternative to a genuine multi-compartment HSPA13.
- **Isoform/construct caveats.** The seed hypothesis rightly asks about isoforms/constructs. Overexpressed or signal-peptide-deleted constructs could mislocalize; none of the primary localization papers here relied on tagged over-expression to claim ER

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P 8131751

used endogenous-scale IF + fractionation). No isoform-specific nuclear/cytosolic localization was found in the reviewed literature.

- **Topology question (genuine).**

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cytosolic Ubl binding to the ATPase core is a real conflict with a strictly ER-luminal model and cannot be dismissed; it is simply unproven by localization.

- **Database carry-over.** The UniProt "plasma membrane" line and the QuickGO IBA set are the database-level source of the hypothesis; they should be treated as orientation, not evidence.

Limitations and Knowledge Gaps

1. **No high-resolution HSPA13-specific imaging.** HPA reports no confident IF localization;

P 8131751

imaging is 1994-era. *What to resolve:* endogenous-tag or validated-antibody confocal/ immuno-EM across ER, nucleus, PM, cytosol.

2. **Cytosolic exposure of the ATPase core is unresolved.**

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shows binding but not localization. *Why it matters:* it is the only mechanistic basis for any cytosol annotation. *Resolve with:* proximity labeling (BioID/APEX at ER lumen vs cytosol), protease-protection topology assays.

3. **NKCC2 contact site not mapped.**

P 33672238

does not localize the STCH–NKCC2 interface to the cytoplasmic tail. *Resolve with:* domain-mapping co-IP / crosslinking to determine whether the contact is luminal or cytosolic.

4. **No proximity-labeling dataset for HSPA13 was directly accessible in this run.** The seed hypothesis references "ER-luminal proximity-labeling evidence"; that specific dataset was not retrievable here and should be located and cited directly by the curator.

5. **Exosome route vs plasma-membrane residence not disentangled.** Exosomal detection

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P 19199708

) does not equal a stable plasma-membrane pool.

Discriminating Tests

1. **Split proximity labeling:** APEX2/TurboID fused to HSPA13 (respecting the signal peptide) plus orthogonal ER-lumen, cytosol, and nuclear baits — directly quantifies which compartments HSPA13 samples.
2. **Protease-protection / topology assay** on microsomes $\pm$ detergent to test whether any HSPA13 ATPase epitope is cytosolically exposed (tests the

P 10675567

implication).

3. **Endogenous CRISPR knock-in tag** (e.g., HaloTag) + confocal and immuno-EM to score nucleus/PM/cytosol pools without over-expression artifacts.
4. **Signal-peptide-deletion construct** to determine whether removing ER targeting redirects HSPA13 to cytosol/nucleus — clarifying whether any secondary pool is physiological or construct-dependent.
5. **NKCC2 interface mapping** (truncation/crosslinking co-IP) to test the "cytoplasmic-tail interaction" claim explicitly.

Proposed Follow-up Experiments / Actions (Curation Leads)

Candidate references + exact snippets to verify:

- P 8131751

— "...*STCH* migrates predominantly as a 60 kDa species and is enriched in a membrane-bound microsome fraction." → supports ER/microsome CC.
- P 18793616

— "...is a microsomal protein that contains a N-terminal ATPase domain but lacks a C-terminal protein binding domain" → supports ER; supports truncated-SBD MF nuance.
- P 33672238

— "...*STCH*...as two novel binding partners of the ER-resident form of *NKCC2*" → ER/ERAD context; **does not** support cytosol.
- P 10675567

— "...ubiquitin-like (*UbL*) proteins that bind to a short peptide within the ATPase domain of the *Hsp70*-like *Stch* protein." → unresolved cytosolic-topology question only.

Candidate GO actions:

- **Retain/strengthen** GO:0005783 (ER, CC) with experimental citation

P 8131751

consider a more specific ER-lumen/microsome term.

- **Flag as non-core / candidate removal** GO:0005634 (nucleus), GO:0005886 (plasma membrane), GO:0005829 (cytosol) — all IBA paralog carry-over, no HSPA13 experimental support.
- **Retain** GO:0070062 (exosome, HDA).
- Consider an **ERAD / ER protein quality-control BP** term supported by

P 33672238

instead of leaving only generic protein binding.

Suggested curator questions: Which specific proximity-labeling dataset underlies the "ER-luminal" claim in the seed? Is any HSPA13 isoform annotated without the signal peptide? Does any experimental (non-IBA) source place HSPA13 in the nucleus/PM/cytosol?

Suggested experiments: the five discriminating tests above, prioritizing split proximity labeling and topology/protease-protection.

Bottom line

HSPA13/STCH is an **ER/microsomal, secretory-pathway HSP70-superfamily chaperone** (signal peptide, no TM, no EEVD, no KDEL, truncated SBD) whose **nucleus, plasma-membrane, and cytosol GO CC annotations are IBA phylogenetic carry-over from cytosolic paralogs with no HSPA13-specific experimental support**. The NKCC2 interaction is an ER-resident/ERAD context and does not demonstrate a cytosolic pool; only cytosolic UbL/BAG6 binding to the ATPase core (a binding assay) leaves a genuine but unproven cytosolic-topology question. **Recommendation: retain/strengthen ER; flag the three secondary pools as paralog carry-over lacking direct support.**